SYM-3: Token-Aware Shared-State Messaging for Agent-to-Agent Communication

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Abstract

Agent-to-agent traffic is billed by tokenizer output, not by human readability. SYM-3 is a family of wire encodings for structured forge lifecycle commands that minimize **o200k_base** token cost under honest decoder setup accounting. Encoders share a tokenizer-tuned lexicon, session-scoped identifiers, optional content-addressed blobs, positional field schemas, default-value omission, and an adaptive mode that picks the cheapest variant per message. On an 11-message synthetic lifecycle, SYM-3 **refs** totals **491** tokens (setup min + messages) versus **682** for compact JSON and **686** for lossless English; savings grow on 40- and 200-message transcripts. Setup tokens are counted in a **decode-sufficient** minimal form (mappings used in the transcript only). Negative results from SYM-2P v1 packets, SYM-2P v2 dense pilots, and multilingual lexicon candidates are reported. Model comprehension on the wire format remains under evaluation.

Motivation

Multi-agent systems exchange structured commands—task claims, artifact hashes, evidence bindings—at scale. When both endpoints are language models or model-backed services, the dominant cost is **tokens on the wire**, not bytes or JSON prettiness. Human operators rarely read raw agent traffic; protocols may use opaque tokens, positional fields, and shared state as long as decoders can reconstruct canonical payloads.

SYM-3 targets **lossless** reconstruction of forge commands relative to compact JSON baselines, while accepting one-time decoder context (lexicon slice + schema) that must be present or cached for decode. Comparisons use tokenizer-accurate counting, not character length or heuristics. The normative profile is **SYM-3/1** (`SPEC-v3.md`); the reference implementation lives under `scripts/protean-sym3/`, with the Synbios forge schema isolated under `schemas/synbios/`.

Prior work

Agora (arXiv:2410.11905) describes hash-referenced, content-addressed agent envelopes; SYM-3 **refs** follows a similar content-store pattern and is the closest prior-art row in our benchmark (11-message total **1027** including Agora setup).

TOON (`toon-format`) provides a compact textual serialization for nested structures; we include a TOON baseline on the same fixtures (**805** message tokens on 11 messages with the repo fallback encoder when the package encoder is unavailable).

LLMLingua / LLMLingua-2 compress natural-language prompts by dropping tokens; they optimize single-turn context, not a **session** with evolving shared state and repeated digests.

Chain of Draft motivates terse natural-language drafts for LLM I/O. We do **not** claim a CoD-style baseline in the bench: the row formerly labeled "terse English (CoD-style)" was renamed to **English (lossless, key=value one-line)**—a lossless `key=value` serialization that tokenizes to **691** on 11 messages, **above** lossless English (**686**), because repeated key names and JSON quoting dominate.

A2A and **MCP** standardize agent discovery and tool/context channels; SYM-3 is complementary wire compression for an application-specific command stream, not a replacement for either protocol.

Multilingual tokenizer fertility: Petrov et al. (2023) show non-English text is not uniformly cheaper under byte-level BPE; our lexicon generator rejected non-English wire tokens that did not reduce **o200k** counts (see negative results).

Method

SYM-3 messages are line-oriented text. A session tracks project, task, agent, and artifact identifiers; repeated UUIDs collapse to single-character wire tokens (`p`, `t`, `a`, …). A **tokenizer-cost lexicon** maps command names, status/kind enums, and format tags to strings chosen by minimum max-token count across configured encodings (primary: **o200k_base**).

State deltas: the delta encoder emits only fields that change relative to session state instead of full canonical payloads.

Content-addressed refs: large or repeated string/blob values hash to stable ref tokens once published in a session content store (Agora-like).

Positional schemas: fixed field order per command reduces key overhead.

Default omission: fields equal to schema defaults are dropped on the wire when safe.

Adaptive encoding: per message, the implementation selects the cheapest among delta, refs, positional, omit, and combined variants (format tag prefix on each line).

Decoder context: decoders need a **decode-sufficient** prefix: wire mappings for every command, status, kind, and format tag **used in the transcript** (for adaptive, all format tags). A **full** setup includes the committed lexicon JSON and complete positional schema; benchmarks report both. **Cached†** totals assume one-time billing of 10% of minimal setup (not API-measured).

Round-trip goal: given decoder context + lines, reconstruct canonical JSON-equivalent payloads (`tests/test_sym3.py` decodes with **minimal context only**). **Decode-sufficient minimal context** is the primary setup column in results. Rule **L-18** (no latent channel) from `SPEC.md` applies: setup is explicit text, not a hidden cache.

Negative results

SYM-2P v1 (this repo codec) on `lifecycle_11`: **2719** message tokens (primary total **2719**) versus compact JSON **682**—SYM-2P JSON packets without shared-state compression are unsuitable for token billing.

SYM-2P v1 (synbios PR #6 pilot reference) on the same transcript shape: **2720** message tokens (recorded as a cross-repo check).

SYM-2P v2 dense (supersedes `SPEC-V2-DRAFT.md` for token efficiency): PR #6 pilot on 11 messages—session steady state **1202** message tokens, handshake setup **213**, total **1415** (still above JSON **683** messages-only in that pilot); **raw v2 dense wire 3386** message tokens without session amortization. v2 remains in the repo as a draft spec and tooling reference, not a migration target for token-aware agent sessions.

Multilingual / non-English lexicon candidates: words chosen for human brevity did not reduce **o200k** token counts versus ASCII wire tokens; the committed lexicon stays tokenizer-tuned.

Results

All numbers below are taken from docs/sym3-bench.md (verified by `make bench` against committed tables). Primary accounting: **setup (min, decode-sufficient)** + **messages**, **o200k_base**.

lifecycle_11 (11 messages)

Format	setup (min)	messages	total (primary)
compact JSON	0	682	682
English (lossless)	0	686	686
English (lossless, key=value one-line)	0	691	691
SYM-3 refs	250	241	**491**

SYM-3 adaptive	271	241	**512**
SYM-3 combined	250	334	584
SYM-3 delta	250	349	599
Agora envelope	48	979	1027
TOON	0	805	805

Under **full** setup (524 + messages), SYM-3 refs/adaptive totals **765**—above compact JSON at 11 messages. Do not compare against English (lossy summary) for information equivalence.

Break-even (messages to beat compact JSON / lossless English): adaptive **7** / **7**; refs **7** / **7**; combined **8** / **8**; delta **9** / **9**; omit **9** / **9**; positional **7** / **7**.

lifecycle_40 (40 messages)

Format	total (primary)	total (cached†)
compact JSON	5343	5343.0
English (lossless)	5242	5242.0
SYM-3 adaptive	**1738**	1490.5
SYM-3 refs	**1717**	1488.4
SYM-3 combined	2566	2337.4

Break-even vs compact JSON: adaptive **3**, refs **3**, combined **4**, delta **6**, omit **6**, positional **4** messages.

lifecycle_200 (200 messages)

Format	total (primary)	total (cached†)
compact JSON	27363	27363.0
English (lossless)	26754	26754.0
SYM-3 adaptive	**7649**	7401.5
SYM-3 refs	**7628**	7399.4
SYM-3 combined	11873	11644.4

Break-even vs compact JSON: adaptive **3**, refs **3**, combined **4**, delta **6**, omit **6**, positional **4** messages.

Verdict

On the shortest transcript, SYM-3 **refs** and **adaptive** beat compact JSON and lossless English under primary accounting. Savings increase with transcript length and repeated content. Cached† totals improve further but rely on the 10% setup assumption.

Model comprehension

> **Status:** pending — do not cite accuracy numbers yet.

Metric	Encoder variants	Result
Decode accuracy (line → JSON)	delta, refs, positional, omit, combined, adaptive	TODO
State-question accuracy (session facts)	all	TODO
Generation round-trip (encode → model decode → re-encode)	all	TODO

Optional harness: `PYTHONPATH=scripts/protean-sym3 python3 scripts/protean-sym3/comprehension.py` (requires API key; skipped in CI).

Limitations

- Savings depend on **repeated content and shared session state**; they scale with conversation length and redundancy, not single-turn prompts.
- **Single-turn inference** cost is unchanged; SYM-3 optimizes multi-message agent sessions.
- Counts use **o200k_base** as primary; other models may differ.
- Lifecycles are **synthetic** forge transcripts, not production traffic mixes.
- **Full** lexicon setup can erase wins on very short sessions (see 11-message full-setup row).
- Human readability and third-party auditability of raw lines are reduced by design.

Reproduction

See docs/REPRODUCE.md. Summary: `make bench` after `pip install -r requirements-bench.txt` (verify-only; use `make bench-write` to refresh golden files).

References

- Agora: arXiv:2410.11905
- Petrov et al., 2023 — tokenizer fertility across languages
- LLMLingua / LLMLingua-2 — prompt token compression
- Chain of Draft — terse drafting for LLM I/O (motivation only; not the key=value baseline)
- TOON — `toon-format` package
- Agent protocols: A2A, MCP (tool/context standards)